

Penrose triangle

The **Penrose triangle**, also known as the **Penrose tribar**, the **impossible tribar**,^[1] or the **impossible triangle**,^[2] is a triangular impossible object, an optical illusion consisting of an object which can be depicted in a perspective drawing. It cannot exist as a solid object in ordinary three-dimensional Euclidean space, although its surface can be embedded isometrically (bent but not stretched) in five-dimensional Euclidean space.^[3] It was first created by the Swedish artist *Oscar Reutersvärd* in 1934.^[4] Independently from Reutersvärd, the triangle was devised and popularized in the 1950s by psychiatrist *Lionel Penrose* and his son, the mathematician and Nobel Prize laureate *Roger Penrose*, who described it as "impossibility in its purest form".^[5] It is featured prominently in the works of artist *M. C. Escher*, whose earlier depictions of impossible objects partly inspired it.

Description

The tribar/triangle appears to be a solid object, made of three straight beams of square cross-section which meet pairwise at right angles at the vertices of the triangle they form. The beams may be broken, forming cubes or cuboids.

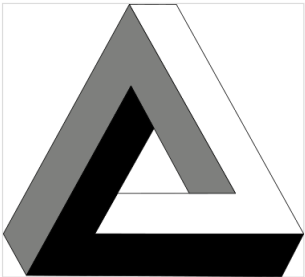
This combination of properties cannot be realized by any three-dimensional object in ordinary Euclidean space. Such an object can exist in certain Euclidean 3-manifolds,^[6] A surface with the same geodesic distances as the depicted surface of the tribar, but without its flat shape and right angles, are to be preserved, can also exist in 5-dimensional Euclidean space, which is the lowest-dimensional Euclidean space within which this surface can be isometrically embedded.^[3] There also exist three-dimensional solid shapes each of which, when viewed from a certain angle, appears the same as the 2-dimensional depiction of the Penrose triangle on this page (such as – for example – the adjacent image depicting a sculpture in *Perth, Australia*). The term "Penrose Triangle" can refer to the 2-dimensional depiction or the impossible object itself.

If a line is traced around the Penrose triangle, a 4-loop *Möbius strip* is formed.^[7]

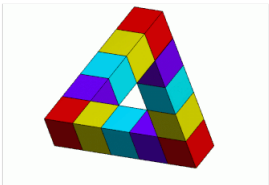
Depictions

M.C. Escher's lithograph *Waterfall* (1961) depicts a watercourse that flows in a zigzag along the long sides of two elongated Penrose triangles, so that it ends up two stories higher than it began. The resulting waterfall, forming the short sides of both triangles, drives a water wheel. Escher points out that in order to keep the wheel turning, some water must occasionally be added to compensate for evaporation. A third Penrose triangle lies between the other two, formed by two segments of waterway and a support tower.^[8]

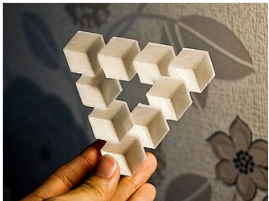
Sculptures



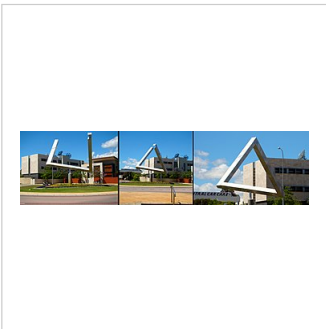
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A rotating Penrose triangle model to show illusion. At the moment of illusion, there appears to be a pair of purple faces (one partially occluded) joined at right angles, but these are actually parallel faces, and the partially occluded face is internal, not external.



A 3D-printed version of the Reutersvärd Triangle illusion



Impossible triangle sculpture as an optical illusion, East Perth, Western Australia



Impossible Triangle sculpture, Gotschuchen, Austria



Real Penrose Triangle, Stainless Steel, by W.A.Stanggaßinger, Wasserburg am Inn, Germany. This type of impossible triangle was first created in 1969 by the Soviet kinetic artist *Vyacheslav Koleichuk*.^[9]

See also

- Impossible trident
- Shepard elephant
- Penrose stairs

References

1.

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5.

Penrose, L. S.; Penrose, R. (February 1958). "Impossible objects: a special type of visual illusion". *British Journal of Psychology*. **49** (1): 31–33. doi:10.1111/j.2044-8295.1958.tb00634.x (https://doi.org/10.1111%2Fj.2044-8295.1958.tb00634.x). PMID 13536303 (https://pubmed.ncbi.nlm.nih.gov/13536303).

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Francis, George K. (1988). "Chapter 4: The impossible tribar". *A Topological Picturebook*. Springer. pp. 65–76. doi:10.1007/978-0-387-68120-7_4 (https://doi.org/10.1007%2F978-0-387-68120-7_4). ISBN 0-387-96426-6. See in particular p. 68, where Francis attributes this observation to John Stillwell.

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External links

- An article about impossible triangle sculpture in Perth (http://im-possible.info/english/articles/real/real3.html)
- Escher for Real constructions (https://web.archive.org/web/20080120055138/http://www.cs.technion.ac.il/~gershon/EscherForReal/)

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